Murmurs of the Silenced: Secure Reporting of Misconduct Settlements

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Abstract

For decades, scholars debated the merits between resolving disputes by public adjudications or private settlements. This tension is particularly relevant in misconduct settlements, where wrongdoers can hide behind the confidentiality available in a private settlement. A paradigmatic example of this was the #MeToo movement and the revelation of serial sexual predators sheltered by secret settlements. Using Multi-Party Computation, a cryptographic technique that enables parties to provide private data for computation without giving up confidentiality, we contribute a fully-interwoven statutory-technological system that implements secure reporting of wrongful misconduct settlements, in order to provide oversight statistics to policymakers and to unmask repeatedly-settling parties for investigation. By providing a unique policy option that balances privacy with oversight, our proposal lessens the need to restrict settlement confidentiality, thereby protecting the autonomy of the parties to settle privately, if they so choose. More broadly, our proposal addresses the oversight complaints against settlements by scholars, and advances the discourse on the appropriate roles of adjudication and settlements in resolving disputes.

CCS Concepts

• **Applied computing** → **Law**; • **Security and privacy** → **Privacy protections**; *Cryptography*; • **Social and professional topics** → **Governmental regulations**.

Keywords

Applied Cryptography, Misconduct Settlement, Data Privacy

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1 Introduction

"My clients signed a confidentiality agreement, [t]hey can't talk to you."[1] Chris and Stephanie Hallowich were property owners in Pennsylvania who suffered headaches, burning eyes, and sore throats, which they attributed to the nearby fracking operation by Range Resources Corp. The family threatened to sue the company, and in 2011, the couple entered into a confidential settlement agreement with Range and other defending parties. The Hallowich family was paid a $750,000 settlement, but gave up their home and their claims in return.

They also gave up the rights to speak for themselves and for their children,[2] then ages seven and ten. Before settling, Stephanie Hallowich was outspoken on the adverse effects of the drilling operations. With the settlement, they agreed to not comment on a full range of topics, including the companies and the broader oil and gas development, never to join future protests or actions against the companies, and signed an affidavit that there is no medical evidence that gas drilling caused their symptoms.[3]

"The matter has been resolved."[4] Nor was Hallowiches' case an isolated encounter with the drilling industry. After Aruba Petroleum began drilling on their property, Tim and Christine Ruggiero, residents of Wise County, Texas, tested a previously-safe well on their land and found that it had chemicals, such as benzene and acetone, that were linked to gas exploration and fracking. They documented their ordeal in a blog post,[5] but shortly after, considered the matter "resolved" and "[will] say no more about it."[6] Records showed that Aruba bought their property for $10 and "other considerations."[7] While the outcome for the Ruggieros is undisclosed, a reasonable

[1] Jim Efstathiou Jr. & Mark Drajem, *Drillers Silence Fracking Claims With Sealed Settlements*, Bloomberg News (Jun. 5, 2013), http://www.bloomberg.com/news/2013-06-06/drillers-silence-fracking-claims-with-sealed-settlements.html.
[2] The Hallowiches' settlement was private and out-of-court, and would not have been known, if not for the presence of minors that required a court's approval. The court ordered the agreement unsealed.
[3] Kelli Fisher, *Communities in the Dark: The Use of State Sunshine Laws to Shed Light on the Fracking Industry*, 40 B.C. Envtl. Aff. L. Rev. 99, 118 (2015); Efstathiou & Drajem, *supra* note 1.
[4] Efstathiou & Drajem, *supra* note 1.
[5] Christine Ruggiero, *When Drilling Starts: Holes in the Ruggiero's Health, Property Values & Landowner Rights* (Jan. 4, 2012), https://earthworks.org/blog/tim_and_christine_ruggiero/.
[6] Tim Ruggiero, *Leaving Gasland* (Jan. 4, 2012), https://earthworks.org/blog/leaving_gasland/.
[7] Efstathiou & Drajem, *supra* note 1.

inference is that an out-of-court settlement was reached: the oil company bought their home, and their silence.

2 Background

Tim Ruggiero might have considered his dispute resolved, but for legal scholars, the troubling aspects of private settlement are very much unresolved. The tension has its roots in the decades-old scholarly debate of adjudication versus settlement. Conventionally, when one party has a dispute against another, they go to court and file a civil complaint. Such a dispute may eventually be adjudicated publicly by a neutral third-party, typically a judge or a jury. A process of discovery and trial ensues, until eventually a judgment is reached, which is imposed on the parties, and serves as guidepost for future dispute resolutions. This judicial process of fact-finding and rule-making animates the common law, and gives occasions for society to render its democratic judgment on emerging controversies.

Alternative dispute resolution ("ADR"), where parties engage in informal, consensus-based processes such as mediation or negotiation to resolve their disputes, gained momentum in the middle of last century.[8] In contrast to public adjudication, private settlements are negotiated between disputing parties. Instead of a judgment imposed on the parties, the settlement is a private contract that both sides agree on. Among the benefits of settling are said to be savings in litigation cost and privacy. By agreeing among themselves and forgoing judicial due process, the parties gain quicker, but private, resolution and the certainty of an outcome that they agree to.

Since being introduced to the courts, the use of settlement has grown considerably, but not without opposition. In response to the increasing popularity and encouragement of settlement to resolve disputes, some legal scholars have voiced strong opposition. One of the earliest critics was Owen Fiss. In his article, *Against Settlements*, Fiss argued against settlement because of imbalances of bargaining power between disputants, questionable assumption of whether disputants truly consent to the settlements, the lack of accountability for wrongdoers, and of judicial oversight.[9]

In the four decades since Fiss published his article, the debate has persisted and evolved, maintaining its relevance today. For instance, a decade after Fiss's critique, Laura Nader argued more broadly that promoting "harmony" through settlement was a form of pacification in the face of class conflicts and demands against ruling interests.[10] David Luban argued more specifically that the problem wasn't settlement per se, but the private nature of some settlements, which removed information from the public purview, essentially removing information that could inform societal values and norms.[11]

Even in the face of such critiques, settlement has proliferated and is now the norm.[12] Most scholarly articles responding to Fiss's critique against settlement argue that there is great public value in settlement. They argue that it is faster and cheaper than going to court and helps alleviate overburdened court dockets.[13] But more importantly, they point to examples where focusing on the relationships between the disputing parties, giving them a chance to communicate, practice conflict resolution skills, and develop an agreement tailored to their needs is critical to the future of a democratic society.[14]

To be fair, many of these articles recognize there are worthy exceptions against settlement, such as where there is a recurring harm, the need for sanctions, or the need to create new law or precedent, but parties have the absolute private right to settle their disputes privately.[15] In 1995, Carrie Menkel-Meadow wrote "Whose Dispute Is It Anyway? A Philosophical and Democratic Defense of Settlement (In Some Cases)" and reframes the debate for or against settlement into a contextual question of when, how, and under what circumstances "should public institutional and structural needs and values override the parties' desire to settle or the court's incentives to promote settlement?"[16]

2.1 Paradigmatic Example: Serial Sexual Predators and Secret Settlements

Sexual misconduct settlements serve as paradigmatic examples of the harm of secret settlement in resolving disputes. In 2017, with the exposure of the Hollywood mogul Harvey Weinstein as a serial sexual predator who has settled many sexual misconduct allegations over a period of several decades with no consequences, the public learned the harm of secret misconduct settlements.[17] A scholarly debate, waged and raged for four decades, between the merits and harm of adjudication vs. settlement, suddenly entered the public discourse. The unequal power dynamics between disputants and the lack of democratic reckoning of sexual misconduct, in the form of public adjudication and judicial oversight, was vividly portrayed by a powerful Hollywood producer and women in less powerful roles. The debate went from academic to visceral.

2.2 Growing Scrutiny of NDAs in Misconduct Settlements

Non-Disclosure Agreements. The concern with the private nature of settlements takes on added weight with the addition of Non-Disclosure Agreements ("NDAs"), which bind the parties to not disclose the details of the settlements or even the existence of the settlement itself. By their nature, settlements are private contracts and are less transparent than public adjudications. Often, parties will add an NDA to an already private settlement agreement, rendering it virtually invisible. Frequently the NDA is added, because the claim is embarrassing or disreputable enough that either, or both, parties would prefer for the existence of the dispute to be shielded in secrecy. For example, in an employment discrimination case, an employer may wish to avoid the reputation of being a workplace that harbors discrimination, while the employee may also be wary of being viewed as a complainer. This secreting away

[8] Owen Fiss, *Against Settlement*, 93 The Yale Law Journal, 1073–1090 (1984).
[9] *Id.*
[10] Laura Nader, *Coercive Harmony: The Political Economy of Legal Model*, Kroeber Anthropological Society Papers (1996).
[11] David Luban, *Settlements and the Erosion of the Public Realm*, 83 Geo. L.J. 2619 (1995)
[12] Yun-chien Chang & Daniel Klerman, *Settlement Around the World: Settlement Rates in the Largest Economies*, 14 Journal of Legal Analysis 80–175 (2022).
[13] Frank E.A. Sander, *Varieties of Dispute Processing*, 70 F.R.D. 111, 131 (1976).
[14] Frank E.A. Sander & Stephen B. Goldberg, *Fitting the Forum to the Fuss: A User-Friendly Guide to Selecting an ADR Procedure*, 10 Negotiation Journal 49–68 (1994).
[15] *Id.*
[16] Carrie Menkel-Meadow, *Whose Dispute is it Anyway?: A Philosophical and Democratic Defense of Settlement (in Some Cases)*, 83 Geo. L.J. 2663 (1995).
[17] Jodi Kantor & Megan Twohey, *Harvey Weinstein Paid Off Sexual Harassment Accusers for Decades*, N.Y. Times (Oct. 5, 2017), https://www.nytimes.com/2017/10/05/us/harvey-weinstein-harassment-allegations.html.

of the dispute and its resolution deprives society of the occasion to render judgment on the appropriate resolution for, or to even be aware of, problematic conducts that are emerging. In cases where the parties agreed on NDAs because of the fear of public judgment, such secrecy can harbor serial settlers, who can hide in the shadow of confidentiality and operate free from oversight of their conducts.

Wrongful Misconduct. Within the universe of disputes that can be settled, one particular concern is the use of private, confidential settlements to resolve disputes between a claimant who has suffered injury and a defendant who have wrongfully harmed the claimant. In such cases, the power and resource dynamics between the parties are often unequal. Confidentiality is frequently part of the settlement, since one or both parties do not desire to publicize the allegation of unflattering behaviors. Hiding behind the secrecy, these confidential settlements prevent larger society's reckoning of wrongful misconduct that may be repeated by one or more defendants.

Table 1 illustrates common scenarios where parties negotiating settlements may have unequal power, and where confidential settlements can mask a pattern of misbehavior from a repeat offender.

2.3 Policy Responses Against NDAs

Just as the rise of settlement engendered a reaction against it, so too was the emergence of NDA. An NDA takes a private settlement and binds the parties to secrecy, rendering it invisible to society. Therefore, the concern with the private nature of settlements takes on added weight with the addition of NDAs. Nowhere is this tension felt more strongly than in the realm of secret sexual misconduct settlements. The story of Harvey Weinstein vividly illustrated how private settlements, made secret by NDAs, enabled a serial sexual predator to prey on women, over decades, with no accountability.

Faced with the sudden realization that settlements with NDAs protected sexual predators and allowed them to repeatedly assault victims without consequences, lawmakers in more than a dozen states have enacted a variety of restrictions on the use of NDAs with harassment settlement in the workplace.[18] These laws come in different forms. Some states, most notably Washington, passed laws that nullified NDAs in any sexual harassment workplace disputes.[19] Without completely banning NDAs, other states selectively regulate NDAs in sexual misconduct settlements, by forbidding the inclusion of NDAs, unless it was the victim's "preference."[20]

Separately, the federal government also enacted tax regime to regulate the use of NDAs in sexual harassment or sexual abuse cases. Specifically, if a settlement is subjected to an NDA, the payer, or the defendant, is not allowed to deduct the cost of settlements or attorney's fees for such settlement. [21]

We highlighted these efforts not as a comprehensive survey, but to illustrate the strong policy desire to protect the right of survivors of sexual misconduct to speak up and animate the social narrative of sexual misbehavior. We note also that while restrictions against NDAs in the sexual misconduct realm represent the most recent and urgent example, they are not the only kind of restrictions that overseers have contemplated against NDAs in settlements. To name non-sexual NDA restrictions: Arkansas statute declares that NDA provisions in settlements concerning "environmental hazard" are against its public policy,[22] while Texas' Rules of Civil Procedure creates a presumption of access for court documents, which extends to confidential settlements that require its courts' approval.[23]

2.4 NDA Regulations Present Wrenching Tradeoffs

The bans against NDAs, or other less severe regulations of NDAs, are well-intended, but there are unintended consequences. The theoretical concern is that restricting NDAs infringes on the freedom of the parties to contract on terms that they find agreeable.

A practical concern is that the ban on NDAs can end up hurting the very victims that the policymakers are trying to protect. Recall that secrecy is a term that the victim can offer to the defendant to incentivize them to settle. The silence of the victim has value, which the defendant is willing to pay for. If the dispute or settlement cannot be hidden, the defendant will be less willing to settle, or would offer less in consideration.[24] In a similar dynamic, the federal government's disallowance on tax deduction for settling sexual misconduct confidentially raises the cost on the settling defendant, and may end up hurting the victim, as the higher real post-tax cost can result in a lower offer to the victim.[25]

Figure 1 illustrates this wrenching policy landscape, when multiple competing policy-goals converge: A general preference for freedom to contract in settlement allows for the use of NDAs to hide settlements from views. But in cases of misconduct, such NDAs are likely to hide the misconduct and provide a safe harbor for serial offenders to escape accountability. In order to eliminate the safe harbors, policymakers attempt to ban or restrict NDAs, but such efforts can harm the victims, by reducing their payoffs or likelihood of securing a settlement.

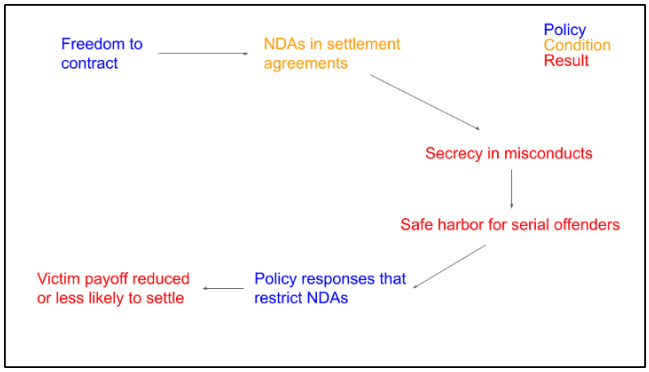


Figure 1: Existing Policy Landscape

[18] Andrea Johnson, Samone Ijoma & Da Hae Kim, *#MeToo Five Years Later: Progress & Pitfalls in State Workplace Anti-Harassment Laws*, NAT'L WOMEN'S LAW CTR. (Oct. 2022), https://perma.cc/4TM5-LGDL.

[19] Johanna Shinners, *Safeguarding Silence: The Weaponization of Nondisclosure Agreements and the Need for More Regulation*, 25 Marq. Benefits & Soc. Welfare L.R. 229, 239 (2024).

[20] N.Y. Gen. Oblig. LAW § 5-336.

[21] 26 U.S.C. § 162(q).

[22] ARK. CODE ANN. § 16-55-122.

[23] TEX. R. CIV. P. 76.

[24] Scott A. Moss, *Illuminating Secrecy: A New Economic Analysis of Confidential Settlements*, 105 Mich. L. Rev. 867, 879-880 (2007).

[25] Gregg D. Polsky, *The Impact of the 2017 Tax Act on Certain Personal Injury Plaintiffs*, Columbia Journal of Tax Law, 12(1), 27–57 (2020).

Misconduct	Victims	Defendants	Pattern of Misbehaviors
Environmental Pollution	residents, landowners	factories, energy firms	An oil company drilling with unsafe practices
Employment Discrimination	workers	employers, firms, businesses	A tech firm hiring exclusively young workers
Housing Discrimination	individuals	landlords, property managers, appraisers	An appraiser undervaluing minority properties
Police Misconduct	residents	police officers, police departments	A police officer prone to excessive use of force
Sexual Misconduct	workers, assistants, social acquaintances	managers, clients, powerful individuals	A sexual predator who preys on interviewees

Table 1: Examples of Misconduct Settlement Parties and Serial Misbehaviors.

A seemingly intractable policy dilemma arises: either tolerate the use of secrecy in misconduct settlements, but know that powerful serial offenders can contract themselves out of accountability. Or, attempt to either ban or regulate confidentiality in misconduct settlements, but with the knowledge that the restrictions would likely harm the settling victims, by lowering compensation or reducing the willingness of the defendant to settle, thus forcing the victims to seek justice by litigation and public adjudication, or give up their claims. This dilemma is particularly wrenching in the case of serial predators with multiple victims, because it is reflecting the tradeoff between the welfare of the current victims, who benefit from strong settlements, and future victims, who benefit from knowing the behaviors of the repeat offenders, in addition to the more effective laws and enforcements enabled by broader societal awareness of such knowledge. We propose a balancing option below.

3 Proposal and Contributions

Responding to the historic debate between adjudication and settlement, made urgent with the legislative actions in restricting NDAs, we propose a policy alternative: an *interwoven*, statutory-technological framework that securely reports secret settlements of misconduct. Using advanced cryptographic techniques only practical since the turn of this century,[26] our solution enables oversight over misconduct, while still permits settlements to be private.

Functionally, we contribute a general framework that scholars and policymakers can readily derive from, to mandate domain-specific misconduct settlement reporting. We envision that lawmakers and regulators can require reporting of settlements in domain-specific contexts, such as employment discrimination or environmental pollution, to a secure reporting registry. The system will then perform two functions: 1) compute useful, domain-specific statistics, and 2) construct an information escrow that detects and unmasks repeatedly-settling parties.

Structurally, we provide a statutory design, a cryptographic design, and connective mechanisms that bind them as a functional whole. To demonstrate scalability, we provide experimental results demonstrating that a basic prototype is able to process hundreds of thousands of settlements, while computing useful statistics and finding repeat settlers.

[26] Dahlia Malkhi, Noam Nisan, Benny Pinkas & Yaron Sella, *Fairplay—a secure two-party computation system*, USENIX Security Symposium (2004).

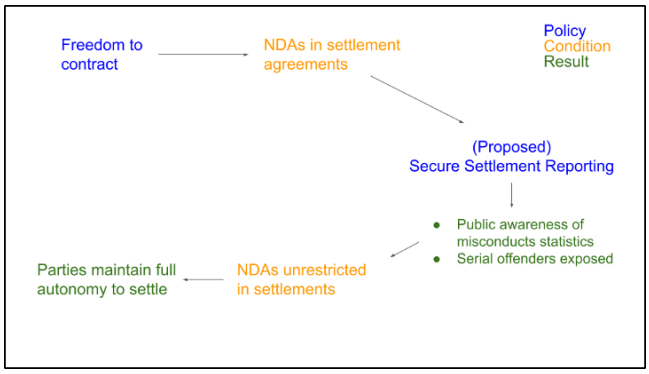


Figure 2: Proposed Policy Landscape

Figure 2 contrasts the policy landscape from Figure 1, when our proposal is introduced. Our option uniquely balances the policy goals, by allowing the freedom for individuals to contract for settlement privacy, while maintaining public awareness and accountability of serial offenses, thereby lessening the need to restrict NDAs and thus maintain settlement potentials for the victims.

Contributions. In summary, our work presents a novel statutory-technological solution that constructs and interweaves essential parts, in both law and computer science, in order to design a new policy trade-off between allowing or banning NDAs in settlements. Pointedly, we contribute:

(1) A legal mechanism that enables a secure reporting registry for domain-specific misconduct settlements
(2) A technical protocol that uses secure multi-party computation ("MPC") to (a) calculate socially desirable and impactful oversight statistics on misconduct prevalence and settlements, and (b) detect parties that settle repeatedly and forward their names to investigators for further review.
(3) A legal mechanism that incentivizes parties to submit data to the MPC protocol by conditioning enforcement of the NDA on submission.
(4) A technical protocol that provides certificates as proof of participation that can be offered as evidence to a judge, arbitrator, or moderator of subsequent dispute between the parties.
(5) A legal mechanism that protects parties who participate in MPC protocol from a claim of breach of NDA or other legal liabilities.

Benefits of Privacy-Enhancing Technologies. Along the way, our construction using cryptography highlights the values of Privacy-Enhancing Technologies ("PETs"), such as MPC, in ensuring *security (confidentiality)* and *accountability* for processing sensitive data. We detail these benefits of adopting MPC in Section 6. Such benefits are not limited to only misconduct settlements reporting. Many other forms of government-required data processing can benefit from adoption of PETs to improve their security and accountability, and our privacy-enhancing cryptographic construction may be of independent interest to policymakers in other areas.

Spotlight on Our Layered and *Interwoven* Approach. To achieve our objective of building a safe and effective settlement registry, we carefully *interweave* ideas from technology and the law. In particular, the legal mechanism in contribution 1 relies on the technical protocol in 2 to protect submission privacy, which relies on the legal mechanism in 3 to incentivize submissions, which relies on the technical protocol in 4 for cryptographic evidence of submission and is protected by the legal mechanism in 5 against disclosure liabilities. The interwoven nature of our design leads to a proposal that is both effective and responsive to stakeholder interests. By contrast, we believe that a tech-only approach is likely to suffer from undue legal risks on input submissions, such as breach of confidentiality or defamation, and unreliable data since there will be no way to enforce accurate submissions; conversely, a law-only approach is likely to face operational risks of data breaches or improper access inherent in large-scale, centralized collection of sensitive data.

The benefits of our proposal are multi-disciplinary. For lawmakers and policy advocates, our solution enables the implementation of settlement reporting registries. Such registries allow the policymakers to relax restrictions on NDA, while providing a flexible level of oversight. For designers of technical protocols and mechanisms, our solution illustrates how specific statutory provisions can combine with a technical construction to both motivate the parties to participate, and to provide legal protection for their participation, thus promoting the adoption of such technologies. For society at large and scholars who are concerned about private resolutions of important disputes, increasing awareness of the prevalence of misconduct enables conversations, deliberations, and ultimately agreement on laws and social norms that address wrongdoings.

3.1 Roadmap

Our work proceeds as follows: We provide an overview of related work in this area. We then provide technical background and highlight scalability results, followed by the actual construction, in the form of statutory provisions and cryptographic constructs. Finally, we provide explanations of our design choices (including the critical role of cryptography) and how they align with our policy goals. We conclude with a vision of a hopeful future.

In Appendix A, we provide a threat assessment of our proposal on both cryptographic and operational levels. In Appendix B, we experimentally validated that our solution can scale into millions of settlements and can be feasibly applied in many domains. In Appendix C (in full version of this work), we describe a hypothetical sexual misconduct settlement registry and demonstrate its use.

3.2 Related Works

Cryptographic Accountability. The use of cryptographic techniques to enable oversight and inform the public has a recent but innovative history. Corrigan-Gibbs and Boneh[27] proposed a cryptographic protocol that can efficiently compute aggregate statistics and preserve privacy of individual submitters. Frankle et al.[28] described a cryptographic protocol that audits cases and accountability commitments of surveillance orders under the Electronic Communications Privacy Act (ECPA), and generalized into "secret information process." Kamara et al.[29] proposed a decentralized, national encrypted gun registry, with a two-layer design that was explicitly responsive to statutory limitation on gun records.

Our work took inspiration from these pioneering works. Like Frankle et al., we strive to provide a protocol that provides accountability while preserving secrecy, and while their work proposed a specific implementation and then generalized, we begin with a framework, and then derive a specific construction of it in the sexual misconduct domain (in Appendix C, included in full version of this work). Kamara et al. described a specific construction of gun registry, designed not only for privacy purposes, but also explicitly to be compliant with federal laws that, in furtherance of political interests, specifically prohibits the storage of individual gun records on the federal level. Our work is similar in that we propose a general framework that can be easily derived to create a domain-specific registry. In keeping with our interdisciplinary focus, we address in detail the policy benefits of our cryptographic registry.

Repeat Settler and Escrow. Daughety and Reinganum provided an economics modeling[30] of the dynamics for settlement considerations between one defendant and two plaintiffs, one early and one later, including the notion of Strongly Correlated Culpability, which captures the scenario of repeat offenders. Their modeling suggests that when the defendant has strongly correlated culpability (a repeat offender), settlement confidentiality will likely have adverse effects on social welfare. Ian Ayres proposed an information escrow,[31] held by an agency such as the U.S. Equal Employment Opportunity Commission ("EEOC"), where in exchange for the enforcement of their NDAs, settlers of sexual misconduct allegations are required to deposit their settlement agreements. This escrow will trigger and release the confidential settlements for investigation, if a party settles repeatedly. Alyson Carrel[32] outlined a

[27] Henry Corrigan-Gibbs & Dan Boneh, *Prio: Private, Robust, and Scalable Computation of Aggregate Statistics*, NSDI (2017).

[28] Jonathan Frankle, Sunoo Park, Daniel Shaar, Shafi Goldwasser & Daniel J. Weitzner, *Practical Accountability of Secret Processes*, USENIX Security Symposium (2018). *See also*, Stephen W. Smith, *Kudzu in the Courthouse: Judgments Made in the Shade*, 3 Federal Courts Law Review 177 (2009), https://ssrn.com/abstract=2080279; Stephen W. Smith, *Gagged, Sealed & Delivered: Reforming ECPA's Secret Docket*, 6 Harvard Law & Policy Review 313 (2012), https://ssrn.com/abstract=2071399 (Judge Smith's articles inspired cryptographic construction in Frankle).

[29] Seny Kamara, Tarik Moataz, Andrew Park & Lucy Qin, *A Decentralized and Encrypted National Gun Registry*, IEEE Symposium on Security & Privacy 1520-1537 (2021).

[30] Andrew Daughety & Jennifer Reinganum, *Informational Externalities in Settlement Bargaining: Confidentiality and Correlated Culpability*, 33 RAND J. ECON. 587 (2002).

[31] Ian Ayres, *Targeting Repeat Offender NDAs*, Stanford L. Rev. Online (Jun. 2018), https://review.law.stanford.edu/wp-content/uploads/sites/3/2018/06/71-Stan.-L.-Rev.-Online-Ayres-1.pdf.

[32] Alyson Carrel, *Reimagining Settlement with Multi-Party Computation*, JTIP Blog (May 16, 2020), https://jtip.law.northwestern.edu/2020/05/19/reimagining-settlement-with-multi-party-computation/.

cryptographic-secure version of Ayres' information escrow, where the submission processing is handled by a cryptographic protocol, instead of by an agency.

Our work is inspired by Ayres and Carrel's proposals, but we expand our scope to include the broad range of misconduct private settlements that can benefit from the escrow-like release to detect repeat settlers. In addition, we also propose to compute beneficial statistics on the prevalence and trends for the collected settlements. Finally, our proposal includes concrete statutory provisions and cryptographic components, interwoven into a framework that has sufficient details to be adopted and made operational by domain-specific parameters.

Secure Allegation Escrow. Kuykendall et al.,[33] Arun et al.,[34] and Koti et al.[35] each proposed different constructions of Secure Allegation Escrows. A Secure Allegation Escrow is a technical device that accepts allegations securely, and reveals the content of the allegation only if multiple matching allegations are submitted. The implementations proposed by each of these prior works differ in terms of their security guarantees and runtime performance.

Our work differs from these prior technical proposals in important ways. We provide a full legal-technological framework, with statutory provisions and technological constructions that have been specifically layered together and designed to operate as a functional whole. More specifically on this point, prior technical constructions have not engaged questions such as how to encourage submissions, nor handle related legal matters such as risk of defamation by the submissions. In contrast, our work provides an enforceability-motivated provision to incentivize parties to submit accurately, a cryptographic matching mechanism that is based on the settlement contract to mitigate false submission, and provide protection against defamation or other disclosure liability, by statutorily excluding the submission as a disclosure.

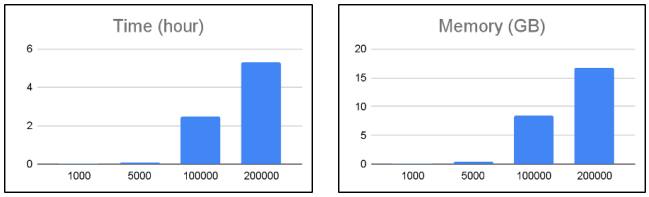
3.3 Technical Background: Secure Multi-Party Computation

Secure Multi-party Computation ("MPC") is a class of advanced cryptographic techniques that allow parties to contribute data for computation while maintaining input privacy. In conventional computing, when parties wish to jointly compute data that they possess individually, their input data must be exposed in order to be computed on. In contrast, with an MPC protocol, two or more parties contribute their data in encoded form, and jointly compute a Functionality, which can be any processing logic. The computation occurs by passing messages between the parties. At the end of the protocol execution, only the result of the computed function is revealed, but the parties' inputs remain private.

[33]Benjamin Kuykendall, Hugo Krawczyk & Tal Rabin, *Cryptography for #MeToo*, PETS (2019).
[34]Venkat Arun, Aniket Kate, Deepak Garg, Peter Druschel & Bobby Bhattacharjee, *Finding safety in numbers with secure allegation escrows*, NDSS (2020).
[35]Nishat Koti, Varsha Bhat Kukkala & Arpita Patra, *Poster: You Share Because We Care: Secure Allegation Escrow System*, IEEE Symposium on Security & Privacy (2022).

3.4 Scalability Highlights

We constructed a general settlement processing prototype, using an off-the-shelf MPC software library, and achieved results of processing hundreds of thousands of settlements in a few hours, using only mid-range consumer hardware.



(a) Run time by settlement count **(b) Memory by settlement count**

Figure 3: Scalability Highlights

Extrapolating from our results, using only high-end consumer hardware, our prototype can easily process more than a million settlements in a day. With dedicated optimization, longer run time, and the use of high-performance computing hardware, settlement count of 50+ millions settlements are feasible. We detail our experimental results and future scalability in Appendix B.

4 A Framework for Secure Settlement Registries

In this section, we describe a framework that can be used to derive domain-specific registries that can accept misconduct settlement submissions and provide useful processing, by outlining the statutory provision that establishes such registries and technical designs to operate them.

4.1 Statutory Framework

To begin, we provide the statutory framework for a reporting mechanism that is built on top of a cryptographically secure protocol. The reporting regime will be authorized by a statute that lawmakers enact, which mandates reporting of a specific kind of misconduct settlement to a registry.

Instead of providing the specific statutory language, we describe in high-level the components needed for the statute to enable such a registry to operate. Our description is *parameterized*. For each statutory component, lawmakers can provide the parameter by drafting language and definitions based on their own jurisdictions and policy needs. We provide comments on how these parameters can be tweaked and what considerations are needed.

We also note that our proposed statutory language is jurisdiction-agnostic. It can be enacted on a state or federal level. While our primary focus is in the United States, our reporting mechanism can also be enacted by countries outside the United States, with minor adoption to local legal systems.

Domain. The statute should define the domain of the misconduct being regulated. This may be a generic category, such as sexual misconduct, or specific causes of action available in the jurisdiction.

Personal Jurisdiction and Coverage. The statute should define the conditions under which a misconduct settlement is subjected to reporting to a registry.

This sets out the conditions that this statutory reporting requirement applies. For example, the statute can base its requirements

on residency, such that if one of the parties is a resident of the state, the resident will need to report the settlement. Alternatively, the statute may require that a settlement contract formed in the state to be reported. The requirement can also be a combination of qualifications. For example, a settlement agreement is required to be reported, if either: 1) one of the parties is a resident at the time of settlement, or 2) if the settlement is governed by the state's law.

The coverage may also vary based on other factors, such as how many years the statute will look back to require submission for each year's computation and data deletion when a settlement has aged beyond the statutory requirements.

Administration. The statute should delegate the regulatory rulemaking and administration of the reporting registry to an administrative agency.

The responsibility could be delegated to an existing agency, such as the California Civil Rights Department or the U.S. Equal Employment Opportunity Commission. Alternatively, a dedicated agency can be created for this purpose.

This administrative agency is tasked with operating a misconduct settlement registry that accepts submissions from the filers. The agency will publish a schema of how settlement agreements should be translated into submission and how to submit. The agency will also specify the precise processing that will be applied to all submissions. More technical details are discussed in Sections 4.2-4.4.

Gap-Filling Responsibility. We stress that this agency will be responsible for filling in gaps in either our framework or the enabling statute. For example, the agency can promulgate rules on what qualifies as submittable misconduct. It can also rulemake to handle edge cases, such as if a party has a name change, lost their private token, or if a settlement comprises non-monetary payout.

Processing Functionality. The statute should mandate the specific processing that will be performed on submitted misconduct settlements in order to gather aggregated statistics from the submitted settlements, such as outputting a count of settlements, the average settlement amount, and a histogram of the settlement amounts. The statute can also specify that if a party has settled more than some threshold of frequency, then their names are subject to be unmasked and forwarded to the appropriate agency or local prosecutor for investigation.

Additional Data Fields. Our construction provides for a base set of data fields to be collected, including settlement dates, parties' names, amount, and other submission-related fields. However, the statute can require additional useful data fields to be collected. For example, the statute can require submission of the postal code of environment pollution, or the industry code where the employment discrimination occurred. These additional data fields can be used by the Processing Functionality to compute more informative statistics.

Canonical Personal ID. The statute should specify that the submitter declare their names on their submission using a name that is on a single kind of ID that can later be verified. Examples of such canonical names include the name on the home country's passport, or a national ID number.

This canonical personal ID will be used to detect parties that have settled multiple times. We stress that this definition must resolve to a single name. This is required to ensure that a party cannot submit a "fake" name or an alias, in an attempt to avoid being unmasked as a repeat settler.

Furthermore, as later described in the enforcement section, this canonical personal ID is part of the requirement of the submission commitment. If a party does not submit their name accurately, the cryptographic commitment will not be verified, and their submissions will be considered void.

Evidence of Submission. The statute should specify that as part of the processing, each submitter should also receive a certificate, with a timestamp, that serves as proof that the settlement has been reported. For judicial economy, the statute can direct courts to take judicial notice[36] on a validly provided certificate as proof of settlement submission in a court proceeding.

Security Requirement. The submission should be accepted using secure multi-party computation, or equivalent cryptographically secure protocols that allows processing without the content of the submission being exposed or stored in plainview.

Any party accepting submissions or computing the secure protocol under the authority of the registry should be required to take proper security measures to secure the storage and processing of submission data. Such parties should not disclose the data, except in accordance with the protocol promulgated by the registry.

This component requires the agency operating the registry to do so securely, so as to avoid the risk of violating filers' privacy after they submitted their settlements. It also obligates parties designated by the agency as compute parties to not disclose, or collude with each other to expose, the submission data.

Enforcement. In order to incentivize parties to submit their settlements and do so accurately, the statute should specify the penalty for if a party does not submit or submits incorrect data. This provision serves as a motivating factor for the parties to submit their settlements to the registry. This is necessary, because such settlements are otherwise private and it would be hard for regulators to discover recalcitrant parties.

As a baseline enforcement incentive, we propose that if an asserting party did not timely and correctly submit their settlement to the registry (including with the correct Canonical Personal ID), they will not be able to enforce the non-disclosure provision of a settlement in a court. By conditioning the enforcement of the NDA to valid submission, the parties will be motivated to submit their settlements correctly to the registry. A jurisdiction may decide on other penalties as is appropriate for the domain and local laws.

Submission Not Considered Disclosure. The statute should declare that, as a matter of public policy, a party's cryptographically-secure submission to the settlement registry is not a disclosure within the meaning of the settlement contract.

This component protects the submitter from being sued over their submission to the registry as a disclosure that breaches confidentiality. Additionally, some settlement agreements explicitly contemplate allowing disclosure that are legally mandated, but require the disclosing party to first give notice and to attempt to avoid the disclosure. By declaring that such cryptographically secure submission is not a disclosure, it protects both the submitter

[36]Judicial notice is an evidentiary mechanism for courts to accept certain facts as undisputed. In this context, directing courts to take judicial notice of the cryptographic proof of submission saves judges from having to rule on whether a cryptographic certificate might be forged, which is a question more proper for cryptographers.

from charge of breach of confidentiality, and it also nullifies any advance-notice or duty-to-object requirement that may be set out in the settlement agreement.

4.2 Establishing the Registry

Technically, a secure computation is realized by a set of compute parties, each of which hold an encoded piece of the inputs.[37] They exchange network messages with each other, and the exchanging of messages constitutes the secure processing of the data.

In preparation for the secure computation, the agency administering the registry should contract with a set of $N >= 2$ compute parties to run the secure computation. Such parties can be any organization, such as law firms, consulting companies, civic organizations, or ADR advocacy organizations such as JAMS[38] and American Arbitration Association. With the use of MPC, these compute parties hold only cryptographic encodings of the data, and cannot access the underlying data that they hold. The only requirements are that each organization 1) has the technical capacity to run an MPC program on a computer that is connected to the network and, 2) is able to maintain a properly-secured database containing its encoding of the submission data. Furthermore, the compute parties should be distinct legal entities and be obligated by the statute not to disclose any encoded data they receive to each other or anyone else, and not to use the encoded data for any purpose beyond the secure registry described in the statute.[39]

Separately, the agency operating the registry should publish a set of regulations governing how the secure computation will be performed. Such regulations include details such as how often the secure computations will be run, what kind of input is required, what kind of processing will be applied to the data, and how the output of the processing function will be made available to the public. More importantly, the agency will also publish the software that individuals can use to submit their settlements, and a separate piece of software that the compute parties will run on their computers to effectuate the secure computations.

4.3 Submitting Settlements

We describe the submission format and submission process below. Where appropriate, we interweave discussion of interpretation, to help readers understand how an actual settlement will be captured by the schema. We note that the agency implementing the registry is free to adopt the format based on its local needs, including collecting additional data such as postal code or attorney bar number, as permitted by the enabling statute.

Interested readers are directed to Appendix C, in the full version of this work, for an example of a submission to a hypothetical sexual misconduct registry using a real-world settlement document.

[37] Somewhat confusingly, these pieces are called "shares" in MPC jargon, but this term of art should be considered as analogous to a stock share–a piece of the whole–rather than as data shared with other entities.

[38] JAMS is the world's largest private alternative dispute solution provider. https://www.jamsadr.com/about/.

[39] MPC alone cannot technologically enforce any of these requirements; we opt for a prescriptive approach because it is unclear how the stated limitations of existing privacy laws and contracts apply when used in MPC. *See* Julissa Milligan Walsh, Mayank Varia, Aloni Cohen, Andrew Sellars & Azer Bestavros, *Multi-Regulation Computing: Examining the Legal and Policy Questions That Arise From Secure Multiparty Computation*, ACM Symposium on Computer Science & Law (2022).

4.3.1 Preparing a settlement tuple. As a first step, a party will prepare a tuple that captures the essential components of the settlement.

$$Settlement = \langle settlement_date, \\ settlement_amount, settlement_key \rangle$$

settlement_date: The effective date of the settlement.

settlement_amount: The amount of the settlement paid out. If there are multiple payers or multiple recipients, this will be the sum of all amounts paid. The amount shall include all cash or cash-equivalent payment such as debt forgiveness or market value of property given. For any payment made in installments, it should include the total amount over all installments, or if the number of installments is unlimited, then it should be ten years of estimated payments from settlement.

settlement_key: The settlement_key is defined as the date of activation, which is the date of the last signature signed, followed by a concatenation of an alphabetical list of the true full legal names of all parties who have rights or obligations under this settlement. The legal full names used must be the names used on the first signature line, or if not present or if the name is an alias, the first true and full legal name mentioned of that party in all the settlement documents, ordered lexicographically.

We note in particular that the activation date is often, but not necessarily, the same as the settlement date. The activation date is the date of the last signed signature. The contract itself may specify a different effective date, but that would not affect the date of the last signature.

We note also that our definition for the names of the parties are specifically designed to deterministically capture a party's true name, and not aliases or variations in name.

We define the settlement key in such a "physical" way to serve an important requirement: the schema must be robustly deterministic without communications. That is, each party, looking at the final copy of the settlement, and without legal research, should arrive at identical settlement keys. This should be so, even if the settlement contract was drafted with unusual legal devices, such as the use of aliases, variants on names, or different ways of calculating effective dates.

This strong determinism is crucial for two reasons. First, when computing the aggregate statistics, the processing function uses the settlement key to deduplicate submissions, so that each settlement is counted only once. If the settlement key is not deterministic, then two parties of the same settlement, submitting separately, may cause it to be counted twice. Second, a schema that is subject to interpretation is also potentially expensive, because lawyers will need to conduct legal research to determine how to file the submission, which in turn drives up the cost of submission.

4.3.2 Preparing a submission tuple. As a second step, the party will prepare a submission tuple that envelops the settlement data. The purpose of the submission tuple is to aid in submission management and to act as a base for proof of submission.

$$Submission = \langle submission_date, submitter_name, \\ agent_name, commitment_token, Settlement \rangle$$

submission_date: The date of the submission.

submitter_name: The legal full name of the party making the submission. It must be exactly the same as the party's full name in the settlement key field, inside the Settlement tuple.

agent_name: The legal full name of the agent or firm assisting the submission.

commitment_token: A commitment_token is a cryptographic commitment that the submitter makes to commit to the settlement_key and the submitter's own canonical personal ID. To generate this cryptographic commitment, the submitter should first generate a secret private token (e.g., a 256-bit string). Then, the submitter can generate a commitment_token value that is a cryptographically-secure hash value of the concatenation of settlement_key, the submitter's canonical personal ID, and the private token.

The submitter should keep the private token secure and not reveal it to an untrusted party. If the submitter is ever called upon to prove that they have made a submission, this private token can be offered as proof of the submission.

Settlement: The settlement tuple previously created.

Once the submission tuple has been prepared, it should be split into N shares using Shamir Secret Sharing.[40] The shares should then be uploaded to each of the compute parties chosen by the agency administering the registry.

4.4 Processing Settlements

Having described the process for gathering settlement submissions, we next describe the computation on the submitted settlement data in high-level terms.

The N compute parties, as chosen by the registry agency, will periodically perform multi-party computation. The processing function that such a computation will perform consists of four phases. For each phase, an output list will be generated.

Validation. First, the processing function will validate the submission. This phase filters out any settlement that may be invalid, such as negative settlement amount, duplicate submission from the same filer, settlement with only one party, or settlement date in the future. The protocol can require the data submitter to assist in this validation process by providing a zero knowledge proof[41] or, for better efficiency, distributed proofs like Prio[42] that the compute parties can verify together. The rejected submissions will have their commitment token published in the output list, along with a reason for the rejection. This way, parties who mistakenly submitted incorrect data can re-submit after correction.

Certificates. In the second phase, the processing function will generate proofs of submission. To do so, the processing function generates a certificate for each submission. In more detail, the submitter provides a cryptographic commitment of the settlement tuple, their canonical personal ID, and a private token, and after this commitment is validated using the process described above, the MPC compute parties jointly create a threshold signature of the commitment. Finally, the list of certificates is outputted.

Unmask Repeat Settlers. In the third phase, the processing function will compute the list of parties that have repeatedly settled more than k times, where k is a threshold set by law. Using the previously validated list of settlements, the processing function will gather a list of all submitters, then maintain a counter for each of them as valid settlements are submitted by them. Once the submitters counting process is done, the function will select those that have settlement count $\geq k$, and unmask the names of the submitter and the filing agent, along with the settlement dates and settlement amounts. Unlike the statistical data gathered in the third phase, this list of repeat settlers should remain private and only be forwarded to the appropriate investigator or local prosecutor for further review. For full secrecy, the processing function can be programmed with the public key of the investigative body, and encrypts the list of repeat settlers with the public key. This way, the list of repeat settlers will only be accessible to the investigators, and not to the compute parties.

Aggregate Statistics. In the final phase, the processing function will compute aggregate statistics on all valid submissions. This phase comprises two steps. In the first step, the processing function must merge and deduplicate all submissions for the same settlement together, using the settlement key in the settlement tuple, which itself is wrapped inside the submission. This step is necessary, in order to avoid having the same settlement counted multiple times, if multiple submissions from the parties are received.

In the second step, the actual statistics are computed. Such statistics include the total number of settlements, a histogram of the amounts of the settlements, and the average settlement amount. The aggregate statistics in this phase is then outputted.

Implications of using MPC for reporting. At this point, we examine in more detail how this work's use of MPC satisfies our policy objectives of protecting the confidentiality of the outputs. Of the outputs generated in the four phases, only the output in the final phase is intended to be open to the public. This is because this information is computed over all statistics, and therefore does not need to be private. However, the outputs in the first three phases are evidence of the submissions from the parties, which need to be private, so that the public is not able to discern who has made submission or what is in the individual settlements.

For this reason, our MPC-based submission process ensures that the two lists of rejected and certified submissions (and all input data that were used to construct them) remain private. This is because they are both based on the commitment token that the filers generate at submission time, which is a cryptographically-anonymous hash value that does not reveal the inputs used to generate it, and are therefore only known to each filer.

- To the public, the output in the rejected list is a list of random-looking lines of text. Only the filers know which entry corresponds to their submission. If they see the entry on the rejected list, they can look up the rejection reason to fix and re-submit.
- In a similar way, the certificate list of valid submission also contains unintelligible lines of text. However, these random-looking texts (or "commitments" in the language of cryptographic design) are actually proofs of submission. If a party to an NDA is called upon to prove that they have submitted, they need only

[40] Adi Shamir, *How to share a secret*, Communications of the ACM, 22(11):612-613 (1979).

[41] *See, e.g.*, Kenneth A. Bamberger, Ran Canetti, Shafi Goldwasser, Rebecca Wexler & Evan Zimmerman, *Verification Dilemmas in Law and the Promise of Zero-Knowledge Proofs*, 37 Berkeley Tech. L. J. 1 (2022).

[42] Corrigan-Gibbs & Boneh, *supra* note 27.

to provide the private token, their submitter name, and the settlement details (i.e., the settlement tuple described above). From the private token, any court or challenger will be able to verify the commitment and be convinced of the party's claim. At the same time, the certificates look random to the general public; without the private tokens, they cannot see who has submitted or the details of the individual settlements.

- Finally, while the list of repeat-settlers will be unmasked at the end of the execution, the presentation can be controlled, or an encryption can be used to seal the output, so that only local investigators have access to them. This is because the repeat settler detection is not itself definite proof of misconduct, but only intended to allow further investigations.

5 Discussion on Registry Design and Operation

Having described the statutory basis and technical specifications for how misconduct settlement registries can operate, we discuss notable features below.

5.1 Data Confidentiality

One notable aspect of our proposed solution is that, at all times, the information contained in the submission is only stored in an encoded, unintelligible form. Only if all of the encoded pieces (or "shares," in MPC jargon) are combined can the input data be reconstructed. In contrast, traditional government data warehousing consists of file cabinets storing papers of submission, or a computer database storing information in plainview. The traditional model enables regulators to have free look at the content of the filing anytime, but it also enables the same to any intruders or leakers.

The use of MPC for processing creates a level of security so high that the agency managing the registry cannot see who has or has not submitted data. This is the reason for the processing function to provide a certificate to each submitter. The agency itself does not store a party's submission. This certificate is necessary to be offered in evidence in court, should a party wish to assert confidentiality claims and need to prove that they have compliantly submitted their settlement to the registry.

5.2 Victim-Friendly Submit-Your-Own Scheme

Our system has been designed to accept a submission from each party, rather than requiring the parties to coordinate and file a single joint submission. The submit-your-own scheme complicates our processing function, because we need to internally match and deduplicate the submissions, in order to not double count the settlements. It also creates potential data discrepancies and chance of a party failing to submit. In contrast, if we require the parties to coordinate and make a single submission, much of the complexity can be avoided.

To explain this deduplication complexity, it is because the information is submitted in a cryptographically-encoded form, the human regulators, in addition to not knowing who has submitted, also cannot tell if two submissions are from the same settlement or two different ones. Only the cryptographically secure processing function has access to the details of the submission to match and deduplicate the submissions. An example submission workflow is provided in Appendix C, in the full version of this work.

We chose the more complicated submit-your-own submission scheme to lower costs for the filers and more importantly, to be friendly to the plaintiffs, who tended to be the parties with less resources and less power. By requiring all parties to coordinate before they jointly file a single submission, it drives up costs for all of them, as the parties need to find each other to prepare the filing. For settlements that were agreed to some time ago, counter-parties may be hard to locate. In addition, requiring the parties to confer and agree to submission gives opportunity for one party to object to submission. This will necessitate negotiation and lawyer billable hours, which increases the cost of filing for all parties. Finally, in cases where the settlement was on conducts that were traumatizing, such as sexual harassment, the need for the parties to coordinate may unnecessarily force plaintiff to reconnect with the perpetrator, therefore risking retraumatization of the survivor.

Our submit-your-own scheme allows each party to submit at their own discretion (unless mandated by the enabling statute). In most misconduct settlements, the defendants almost always prefer confidentiality. They are also typically the ones with more wealth and resources. Therefore, defendants tend to be ones with greater incentive to submit and great ability to bear such costs. In contrast, with our scheme, a plaintiff may opt to not file a submission, whether for emotional comfort or financial reasons, and the penalty of not being able to enforce the NDA will not likely affect them adversely.

6 The Critical Role of Cryptography

At first glance, our proposal does not seem to depend on cryptography. Governments have long required reporting in various realms, from income tax return filing to sex offender registry. Such filings are usually done by paper or electronic filing process. Even if secure channels are used, the information reported is received as plaintext and visible to the collecting agencies. Once received, they are stored in access-controlled warehouses or electronic databases. No advanced cryptography is required in such collections.

We argue that the use of MPC critically accomplishes two goals when processing sensitive data: security and accountability.

6.1 Security

Broadly, we are proposing a general framework for reporting various kinds of misconduct settlements. This means that we are asking citizens to submit sensitive information about their conducts to the government. A centralized depository of such information presents a significant security risk in a democratic society. In the modern era, data breaches have become commonplace.[43] Confidential information about citizens, politicians, and public figures enter the national conversation because of external hackers or internal leakers. Such data breach can be inflammatory and harmful, as when actors breached the DNC email servers and leaked embarrassing, but not wrongful, emails to the public, which nonetheless had an effect on the election.[44] Consider our example in Appendix C (in

[43] Devlin Barrett et al., *U.S. Suspects Hackers in China Breached About 4 Million People's Records, Officials Say*, Wall St. J. (Jun. 5, 2015), https://www.wsj.com/articles/u-s-suspects-hackers-in-china-behind-government-data-breach-sources-say-1433451888.

[44] *Assessing Russian Activities and Intentions in Recent US Elections* (Jan. 6, 2017), https://s3.documentcloud.org/documents/3254237/Russia-Hack-Report.pdf.

the full version of this work) of a sexual misconduct settlement registry—the registry would hold scandalous information on some of the most powerful and public figures. If such information is leaked, it would cause great and irreparable damage to not only the reputation of individuals, but also security of the nation.

Settlements of misconduct are not convictions. A mandatory settlement reporting regime would ask many citizens, under allegations but unconvicted of misconduct, to accurately testify that they have paid compensations to settle unflattering accusations. This depository of information would be a great source of vulnerability and a target for foreign actors. Considering the repository's value and the capability of the actors that would be interested in breaching it, standard access controls would be insufficient. If the existence of their settlements is leaked, these citizens may well be convicted in the court of public opinion. With such grave risk on the line, while we advocate for reporting as scholars, it is imperative for us to seriously engage the security aspect of our proposal.

Considering the security implications, we argue that the role of cryptography in our proposal is not merely an add-on, but a critical component. Advanced cryptography, such as MPC, allows us to be highly confident that the sensitive information, which we advocate reporting of, to be strongly protected from both disgruntled or ideological insiders, or external actors driven by commercial or geopolitical interests. An intriguing question for future research is how to design settlement registries that ensure that an individual's contributed data is cryptographically deniable and can only be revealed by consent.

6.2 Accountability and Transparency in Data Processing

Equally important, we argue that the use of MPC embodies the accountability inherent in our proposal. Observe that when policymakers design a secure MPC protocol, a specific functionality is defined. This functionality is programmed to only process data in a way that the parties agree on (we intentionally use the word "parties" to mean *both* democratic stakeholders and MPC compute parties). It cannot be used to process data in ways that the parties do not consent to. In other words, if citizens are required to submit their settlements of misconduct, they should be confident that the data will only be processed as they expect. For example, if a statute is democratically enacted to require submission of employment discrimination and the functionality is defined to compute only the average settlement amount, it should not later be arbitrarily changed to compute some other values such as the highest settlement amount, or to expose parties' names. This is in contrast to traditional government data collection and processing—once collected, it would be just as easy for the bureaucrats to compute the highest amount or to inspect all submitters' names, as it was to compute the original goal of average amount.

We admit that the MPC function definition can also change. Lawmakers can later amend the statute and specify new computations on the submitted data, even if such changes deviate from the expectation embedded in the earlier submissions. However, unlike traditional government data processing, where accountability can be casually violated (even internal access control depends on adherence[45]), changes to the MPC function must be explicitly authorized and programmed, thus providing a cryptographic measure of data processing accountability and transparency. A fruitful area for future research is on ways to further guarantee cryptographically that submissions are only used for computations specified by law.

7 Benefits Provided by Settlement Registry MPC Outputs

Our interwoven approach delivers several policy benefits: the availability of aggregate statistics enables greater settlement efficiency, enhanced potential for narrative correction, and leads for accountability journalism, while the ability of the protocol to unmask repeat settlers can aid investigators to timely expose serial offenders. Below, we detail how policy and investigative needs can be served by outputs from such cryptographically-secure settlement registries.

7.1 Informing Decisions by Aggregate Statistics

At first glance, one might question the utility of having only statistical data about the total number of settlements, the average settlement amount, and a histogram of settlement amounts as compared to, say, the details that would be available in a complaint or in a judicial ruling if the disputes were taken to court for public adjudication. However, even a simple set of statistics, with the additions of some domain-specific context statistics such as industry or geographical postal code, can be powerful sources of information for both individual disputants and public monitors.

Settlement Efficiency. Settlement statistics can improve settlement efficiency, by speeding up the initial negotiation of future settlement negotiations, and serving as a guidepost for the eventual settlement amount. Knowing the range of settlement amounts gives the parties a short ramp up to substantive discussion, by helping the parties to skip past the initial stage of sky-high demand from the plaintiff (e.g., millions of dollars) and too-low "nuisance-value" offer (e.g., a few thousand dollars) from the defendant.[46] As negotiation progresses, the histogram of amounts can guide the parties to the proper settlement amount, given the severity of the conducts that they agree on.

Narrative Correction. Availability of settlement statistics helps to correct society's perceptions about particular kinds of misconduct. As Minna Kotkin argued, settlement outcome matters, "[f]or the same reasons that verdicts matter: They affect public perceptions and public policy."[47] Koltin was writing about employment discrimination, but the same narrative corrective potential exists for other kinds of misconduct, whether it be environmental pollution or housing discrimination. Indeed, the sudden reckoning of #MeToo can be viewed as a multi-decade failure to correct the narrative of sexual misconduct. More generally, Judith Resnik and Stephen Yeazell both called for better availability of settlement data,[48] and

[45] *See, e.g.*, Eileen Sullivan, *Former Contractor Who Leaked Trump's Tax Returns Sentenced to 5 Years in Prison*, N.Y. Times (Jan. 29, 2024), https://www.nytimes.com/2024/01/29/us/politics/irs-trump-taxes-prison.html.

[46] Moss, *supra* note 24, at 899-900.

[47] Minna Kotkin, *Outing Outcomes: An Empirical Study of Confidential Employment Discrimination Settlements*, 64 Wash & Lee L. Rev. 111, 113 (2007).

[48] Judith Resnik, *Uncovering, Disclosing, and Discovering How the Public Dimensions of Court-Based Processes Are at Risk*, 81 CHI.-KENT L. REV. 521, 534-35 (2006); Stephen C.

our proposal for cryptographically-secure settlement registries answers their calls for better data on misconduct settlements.

Lead for Journalists. Availability of settlement statistics can serve as leads for accountability journalism. Roy Shapira has argued that the legal system, with public dockets that house complaints, pleading, ruling, and other background information, provides a form of "information subsidies" for accountability journalism,[49] by giving them a source to understand an emerging problem and to investigate journalistically. The publications of settlement data can serve a similar function, by giving accountability journalists a source of investigative leads. For example, if employment discrimination settlement data is available, journalists can focus on the industries that settle discrimination claims frequently. While individual settled parties may not be able to discuss in detail, a journalist can leverage the statistics to focus their investigation, much as Jodi Kantor and Megan Twohey broke the story on Harvey Weinstein's history of secret sexual assault settlements.[50]

7.2 Investigative Efficiency and Exposing Serial Offenders to Sunlight

In addition to computing useful statistics, we propose that our functionality unmasks the identities of parties who repeatedly settle and forward them to investigators. This functionality enables regulators and policymakers to identify and focus on those entities most likely to engage in repeated harmful behavior. The act of repeatedly settling misconduct could be a strong indication that the settler is a serial offender who is using settlement as a way to avoid accountability and continue misbehaving without oversight. The paradigmatic example of this is the story of Harvey Weinstein, whose use of secret sexual assault settlements with his victims shielded him from accountability for three decade.[51] Had a secure sexual misconduct settlement registry been in place, Weinstein's pattern of repeated settlements *could have triggered an investigation years earlier*, potentially protecting later victims.

The availability of the names of parties who settle repeatedly is a significant investigative lead that can help regulators focus on the most troubling actors. In the case of oil and fracking companies settling cases confidentially, if investigators receive a list of companies that settle repeatedly, they can focus on the setters' operations to determine if any illegal environmental pollution occurred. To give further context to the benefit of allowing an agency to focus on serial offenders, consider the statistics available at the EEOC: In 2023, more than 81,000 cases were filed, and the agency had a backlog of more than 51,000 pending matters, with an average waiting time of ten months for resolution. Further, on matters that could not be resolved on the agency level, only 143 cases were taken to federal courts.[52] These numbers suggest an agency that is overwhelmed to process all the complaints filed with it.[53] A settlement registry that accepts submissions and unmask repeat settlers can serve two functions: 1) allow the agency to focus on the most notorious settlers that have repeatedly triggered allegations of misconduct, and 2) by stopping serial offenders early in their tracks, less cases involving the same offenders will be filed in the future, thus lightening future case load.

We stress that the list of unmasked repeat settlers is not intended for public release. Not all serial settlers are serial offenders. We recognize that there may be innocent explanations for why a party has settled multiple misconduct disputes. The list of repeat settlers is therefore intended to be forwarded confidentially to regulators or prosecutors, who can manually review the facts and decide if enforcement is warranted.

8 Conclusion

Motivated by the long-running debate between the benefits and drawbacks of private settlements, and particularly focused on the often secret settlements in misconduct disputes, notably exemplified by the revelation of serial sexual settlers-predators in the #MeToo movement, we proposed a joint statutory-technological framework for creating cryptographically-secure reporting registries that accept misconduct settlement submissions and process useful computation on the cryptographically-submitted data. The computations that we proposed include aggregate statistics for oversight and escrow functionality for referring serial settlers to investigators.

More broadly, by proposing a solution that uniquely balances the autonomy and privacy of settling parties with the need for public oversight, we advance scholarly discourse on the appropriate role of settlements and policy considerations of regulations on NDAs, and show how we can provide both strong settlement potential for current victims, yet maintain public oversight to protect future victims. In the end, our proposal suggests a hopeful world where, even as individual settlements are silent, their collective voices are still heard, and democratic reckoning of misconduct is possible.

Acknowledgments

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Yeazell, *Getting What We Asked For, Getting What We Paid For, and Not Liking What We Got: The Vanishing Civil Trial*, 1 J. EMPIRICAL LEGAL STUD. 943, 966-71 (2004).

[49] Roy Shapira, *Law as Source: How the Legal System Facilitates Investigative Journalism*, 37 Yale L. & Pol'y Rev. 153, 158 (2018).

[50] Rose Minutaglio, *How These Two Women Finally Exposed Harvey Weinstein* (Oct. 23, 2017), https://www.marieclaire.com/career-advice/a13051614/harvey-weinstein-jodi-kantor-megan-twohey/.

[51] Kantor & Twohey, *supra* note 17.

[52] *2023 Annual Performance Report* (Feb. 23, 2024), https://www.eeoc.gov/2023-annual-performance-report#_Toc150772729; *What You Can Expect After You File a Charge*, https://www.eeoc.gov/what-you-can-expect-after-you-file-charge (last visited Oct. 1, 2024).

[53] Minna Kotkin, *Reconsidering Confidential Settlements in the #MeToo Era*, 54 U.S.F. L. Rev. 517, 528 (2020).

A Threat Model

Building upon the discussion of data confidentiality in the main body, this section describes potential threats in the design of our registry, and ways to mitigate those threats.

A.1 Long-term Security of Persistent Submission Data

In our system, we envision that the submission, once made, will remain active in the system across multiple rounds of processing. This persistence of the submitted data is a great convenience to the parties, who can submit once and do not need to re-submit, unless the registry amends the submission format to require more data.

However, we point out that by having the data persist in the databases of the compute parties, even in a cryptographically encoded fashion, the risk remains that as time goes on, an external attacker may incrementally penetrate the computers of more of the compute parties. If an attacker can gather enough of the encoded pieces (or "shares") of the data, they can recombine the data and gain access to all the submissions that have been made. Such a full range of compromise would negate the guarantee provided by cryptographically secure multi-party computation. This is not an idle possibility-A depository of private sexual misconduct settlements is a hugely valuable pile of information, and attackers would be well-motivated to persistently attempt to compromise all shares of the submitted settlements.

We envision that this weakness can be mitigated by several methods. First, when the registry is established, numerous compute parties should be chosen. This will ensure that any attackers must successfully attack a large number of storage parties before they can gain access to the data. Second, we recommend that the selected compute parties come from multiple industries or fields, ideally with different operating environments. This heterogeneity will make it difficult for an attacker to compromise all the compute parties with just a single method. Finally, in addition to the use of MPC, we recommend that all compute parties adopt computer security best practices such as regular audits, deployment of firewalls and intrusion detection systems, and isolating various system components to only essential personnel. This defense in-depth strategy will make it difficult for any attacker, even ones well-motivated, to compromise all the compute parties and thus the registry itself.

A.2 Risk of Compromise of Private Token

We now switch gears from security considerations of the compute parties to those of the claimants and defendants who contribute data into the MPC system. The presence of the private token is a critical component in the security for an individual submission. Most importantly, the private token is used in computing the certificates that will be used as proof of submission to the registry. Anyone in possession of the private token, or is able to determine it, will be able to compromise the confidentiality of the submission system, by proving that a certain settlement of parties have submitted.

For example, if Alice and Bob settled a sexual misconduct complaint, and they submit their settlements to the registry, a malicious actor can reveal their submission by publishing their private tokens. Once published, any member of the public can then generate the same certificate and match it to the MPC processing output, thereby obtaining proof of existence of a settlement between Alice and Bob.

Because of the critical nature of the private token, we specify that it should be a cryptographically generated random string of a sufficient length, such that it should be computationally infeasible for an attacker to guess by brute force what the private token is. We suggest a string that is at least 256 bits long,[54] since it contains more numbers than the number of particles on earth.[55]

Besides guessing by brute force, the privacy of the submission could be compromised, if the submitting party has poor security practices and has their private string stolen by conventional means such as physical intrusion. However, because the certificate for each submission is computed independently from its own private string, the compromise of one settlement submission will not affect the confidentiality of the other submissions in the registry.

We remark that additional cryptographic protections can be added to our framework in order to protect victims and defendants who lose control of their private tokens, if there is a public key infrastructure. First, the work of Frankle et al.[56], USENIX Security Symposium (2018). can be added to our framework in order to ensure that only a judge could verify any property of a private token. Second, it is also possible to augment our framework to provide a cryptographic guarantee called "deniability," so that even if the public key infrastructure is compromised, still nobody outside of the judicial system could prove that a private token corresponds to a submitted entry into the MPC system.[57]

A.3 Invalid or Malicious Submission

The fidelity of a secure settlement registry could be degraded, if parties make invalid or malicious submissions. Such submissions could happen by negligence or by intention. For example, a careless party may submit a settlement with the wrong date or the wrong settlement amount. Alternatively, it is conceivable that some external actors may attempt to "flood" the registry with malicious submissions that do not correspond to actual settlement. Such an attempt may skew the computations of the settlement amount average or the total number of submissions received. In particular, specifically-crafted submissions could trigger improper unmasking of targeted parties.

An effective way to mitigate such invalid or malicious submissions is to require the submissions be filed by a licensed lawyer in the state, and to trace the submission to the lawyer's identifier, such as a public key or a bar number. The requirement of a submission by a licensed lawyer has several advantages. First, the lawyer can validate the settlement document and assist in correctly filing the submission. Second, if someone tries to "flood" the registry, the lawyer can note the presence of the client and decline to file multiple submissions. Finally, because the processing function is able to match settlements and calculate statistics only from matched settlements, it further raises the difficulty for an attacker to make matching submissions.

[54] 64 alphanumeric characters in base 16 encoding, or about one line of text.
[55] Edmund Whittaker, *Eddington's theory of the constants of nature*, Math. Gaz. 29(286), 137–144 (1945).
[56] Frankle et al., *supra* note 28.
[57] *See, e.g.*, Rawane Issa, Nicolas Alhaddad & Mayank Varia, *Hecate: Abuse Reporting in Secure Messengers with Sealed Sender*, USENIX Security Symposium (2022).

We are mindful that a requirement for submissions to be filed by lawyers could impose costs on parties to such settlements who are not well-resourced, which are often the claimants. We envision that this access to legal service issue could be resolved by making the submission a service that qualifies for pro bono work that lawyers often do.

Additionally, in the processing function, the validation phase can be enhanced to detect invalid submissions. For example, the validation can be coded to detect settlement dates or amounts that are out of range. It can also aggregate the submission by the lawyer or firm that is filing the submission. If a firm appears to submit an excessive amount of submission, such submissions can be rejected.

B Experimental Results

The use of Multi-party Computation as an advanced technique for privacy-enhanced computation also comes with significant overhead. Such overhead limits the feasibility of processing large amounts of data and/or performing highly-complex computation.

To ensure that our use of MPC to process settlement data in a privacy-preserving manner is feasible in real-world domains, we performed experiments and demonstrated that the current level of technical advances allow useful computations on a meaningfully large amount of settlement data.

Our implementation is intentionally simple. The total engineering effort, including data generation and logic programming, but not including experimental run time, was less than 50 hours.

B.1 Design Requirements

For each scalability level, we assume that N settlements are present, each settlement having two parties. Each party submits the settlement independently with no coordination, for a total of $2N$ submissions. For simplification, we assume settlements always have matching pairs of submissions.

Each submission consists of a number of data fields, including a settlement date (2 bytes), the two names of the two parties (40 bytes each), the amount of settlement (4 bytes), the settlement year (1 byte), the submitter's name (20 bytes), and a proof-of-submission commitment token (32 bytes).

For our experimental scenario, we compute for three kinds of output: 1) a statistical aggregation to calculate the number of settlements, average settlement amount, histogram of the settlement amounts, and yearly count of the settlement years; 2) an unmasked list of parties that are involved in three or more settlements; and 3) a list of proofs in the form of cryptographically-anonymous commitment tokens, usable later to prove submissions.

B.2 Implementation Details

For sample data generation, we created[58] a python script that can generate N settlements, including the intentional inclusion of several repeated parties. The script further secret-shares the resulting data into two files, suitable for computation by 2PC, and outputs the resulting bits into two files, a and b.

[58]Our implementation can be reviewed at https://github.com/ckpeter/secure-settlement-reporting/.

For actual processing, we developed a C++ program that utilizes the EMP-Toolkit library[59] for 2PC MPC implementation. The program reads in the appropriate share of the bits (a for Alice and b for Bob party), securely reconstruct the original submission values, sort the submissions items, then mark any duplicate submissions to avoid double-counting the settlements. The program also records the parties to each settlement and tracks parties that occurred more than a predefined number of times. Finally, the program outputs the submitter-provided commitment tokens, which are anonymous tokens that can be used by the submitter to prove their submission of a settlement with specific attributes.

B.3 Execution Environment

We ran the experiments on Google Cloud Platform ("GCP") using the C2D family of instances, which is a family of instances using the AMD Milan family of CPUs that became available in February 2022.[60] Our experimental nodes run in the same GCP region and same zone (us-central-1b), connected with a minimum of 10 Gbps network interface. We observed ping time generally between 0.15 ms - 0.35 ms. Because we use VM-VM same-zone connection with internal IP, there is no cost for the network traffic.

Our instances were configured with 2 CPUs and 16 GB of RAM (cost: \$0.1225 USD/hr),[61] except on $N > 100$K, we used instances with 4 CPUs and 32 GB of RAM (cost: \$0.2449 USD/hr). Our storage layer is the GCP pd-standard disk with size 100 GB. Our testing revealed a typical read throughput of 101-120 MB/s. Random storage IO performance is not a major consideration, as our program only reads in an input file sequentially. The 100 GB disk costs \$4/mo. All prices are on-demand, in USD.

B.4 Benchmark Results

We chose to test settlement counts of up to 200,000, which in practice would run in a few hours on a mid-range laptop of today. For each size count, to demonstrate feasibility and avoid performance fluctuation in a cloud environment, we performed a minimum of three runs and used the best number. Table 2 shows the time, traffic, memory, gate count, data size,[62] and cost for each settlement count.

B.5 Future Scalability

Our implementation is intended as a demonstration, and as such, has limited optimizations. Our simplifications included: 1) the algorithm uses a simple sort on a single thread, 2) being hosted on a shared-tenant cloud platform that is subjected to variability in CPU performance and network connectivity, and 3) a processing time limit of a few hours.

Additionally, we limited the dataset to what can be processed in a few hours on a mid-range laptop. With a higher end configuration

[59]Xiao Wang et al., *EMP-toolkit: Efficient MultiParty computation toolkit*, https://github.com/emp-toolkit/emp-tool.

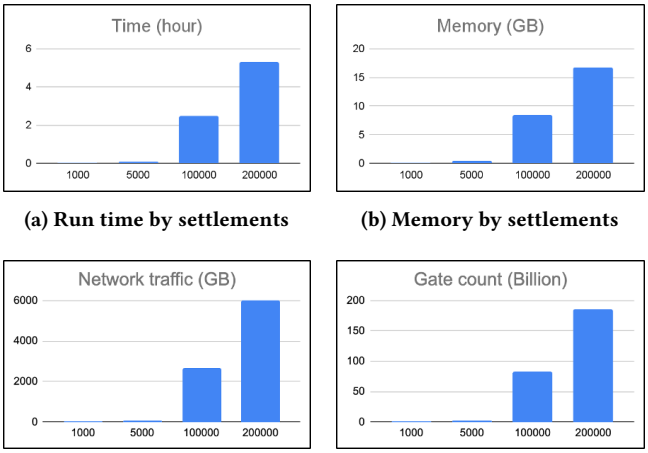
[60]*Introducing Compute Optimized VMs powered by AMD EPYC processors* (Feb. 10, 2022), https://cloud.google.com/blog/products/compute/introducing-compute-optimized-vms-on-amd-epyc-milan.

[61]*VM instance pricing*, https://cloud.google.com/compute/vm-instance-pricing#compute-optimized_machine_types (last visited Oct. 1, 2024).

[62]Data size encompassed all experimental data files, including raw settlement data, settlement data in bits format, and the two secret-shared copies of the raw bits. In actual processing, only a single copy of the secretly shared data, or about 1/3 of total size, would be present on and used by each processing node.

N	Time (best)	Network traffic	Memory	Gate count	Data size	2PC $/run
1k	59 sec	12 GB	95 MB	362 M	13 MB	> $0.01
5k	368 sec	80 GB	437 MB	2491 M	62 MB	$0.025
100k	8904 sec (2.5 hrs)	2660 GB	8567 MB	85390 M	939 MB	$0.6
200k	19075 sec (5.3 hrs)	6000 GB	17198 MB	189780 M	1877 MB	$2.60

Table 2: Benchmark Results



(a) Run time by settlements **(b) Memory by settlements**

(c) Traffic by settlements **(d) Gate count by settlements**

Figure 4: Run time, memory, gate count, and data size by settlement count.

that is still within today's prosumer hardware, and a run time of a day, a million settlements is a reasonable lower bound of what can be processed.

We outline several limitations of our implementation that can be relaxed straightforwardly, using only today's cryptographic and hardware advancements:

Algorithmic improvements. Our current implementation is single-thread only. However, a significant amount of processing is spent on sorting the submission securely. We used bitonic merge sort as the underlying sorting algorithm for the processing, which can be parallelized if multiple processors are available. With today's processors available with 32 or more cores, parallelization may further speed up the processing by 5x or more.

Additionally, there are likely domain-specific optimizations that can be made to further reduce processing time. For example, if each party can be identified by a national or taxpayer ID, then instead of processing 40 bytes for each name, it may be possible to process only 4-8 bytes for an ID. This could meaningfully reduce the amount of data computed, and thus the processing time.

Better hardware. Our testing was performed on a public cloud using a 2-year old generation of hardware (2022). With near-term hardware advancements in both processing and networking speed, it would be reasonable to assume that a 2x increase in processing performance can be achieved.

Longer run time. Many of the statistical gathering and processing can be run on a quarterly basis, which means that an actual run time of 10-30 days would be reasonable. Therefore, even under our simplified implementation, datasets of up to 10-30x larger could be processed under a longer processing period.

In conclusion, our experimental results demonstrate that our proposed solution can easily accommodate a dataset of more than a million settlements. With algorithmic improvements, better hardware, and a longer run time, upwards of 50 million settlements could be reasonably processed.

B.6 Domain Application Feasibility

To contextualize the utility of being able to scale to processing 100+ millions settlements, a few statistics are instructive:

Partnership tax returns. In 2023, the US tax administrative agency, the Internal Revenue Services, received about 5 millions returns on partnerships.[63] Each partnership is a business that consists of multiple partners in a business relationship, and each partner is required to file their own partnership return each year. This is not unlike a settlement under our proposed solution, where each settlement consists of two or more parties, and each party would be expected to file their own returns.

If we view our settlement reporting system as a way to model processing each partnership relationship and/or to detect partners that are present in multiple partnerships, our implementation can comfortably process all 35 million partnership tax returns, over a 7-year period, in the US.

Sexual misconduct. True scale of sexual misconduct statistics is hard to come by, because of its nature and that settlements would commonly be confidential. However, using statistics[64] from the U.S. Equal Employment Opportunity Commission, an agency tasked with regulating workspace discrimination, between FY 2014 - FY 2021, there are annually between approximately 5500 to 8000 complaints received regarding sexual harassment in the employment context. With an expansive assumption of every complaint being valid and resulting in a settlement, this could be 8000 settlements x 7 years = 56,000 settlements within a 7 year lookback period.

Assuming that only 1 out of 10 actual settleable cases of workplace sexual misconduct is complained to the EEOC, or about 560,000 possible settlements within a 7-year period, the volume of data to process for a Workplace Sexual Misconduct Registry is easily within the million-plus scalability of our implementation.

B.7 AI Use Disclosure

Generative AI was used to create the initial boilerplate python data generation script, C++ parsing code, reverseBits(), and shell script to monitor memory and network usages. Github Copilot was used for auto-completion, with manual review for correctness. Actual processing logic, including the use of EMP library, statistical calculations, and repeated settler detection, was coded manually.

[63] *SOI Tax Stats - IRS Data Book index of tables*, https://www.irs.gov/statistics/soi-tax-stats-irs-data-book-index-of-tables (last visited Oct. 1, 2024).
[64] *Sexual Harassment in Our Nation's Workplaces* (Apr. 2022), https://www.eeoc.gov/data/sexual-harassment-our-nations-workplaces.